

RIDSS TEAM MANAGER MODULE

STATUS: 100% PASSED
10 / 10 VERIFIED

COMPREHENSIVE TEST REPORT

System: RIDSS Engine | Module: Team Manager (/api/v1/team-manager) | Execution Date: Sep 28, 2026 | Environment: Async FastAPI + SQLite + FastF1

1. EXECUTIVE SUMMARY

Total Executed	Passed	Failed	Pass Rate	Security Audit	Health Blocking
10	10	0	100%	VERIFIED	VERIFIED

2. AUTOMATED TEST MATRIX & RESULTS

Test ID	Feature Area	Test Description	Actual Result Details	Status
TC-TM-001	Manager & Team Resolution	Validate Team Manager user resolution and team_id scoping	Manager: Daril Tom Jose Team: Oracle Red Bull Racing	PASSED
TC-TM-002	Dashboard Summary & Audit	Verify GET /dashboard metrics & AuditLog feed integration	5 drivers, 2 vehicles, 1 active pairing, 15 audit items	PASSED
TC-TM-003	Driver Roster Management	Verify GET /drivers listing and PATCH /drivers/{id} update & audit	Updated & restored driver #22 (Yuki Tsunoda) details cleanly	PASSED
TC-TM-004	Staff Directory & Tenure	Verify GET /staff and PATCH /staff/{id} user.team_since management	12 staff members fetched (admins excluded). Updated tenure	PASSED
TC-TM-005	Vehicle Inventory & Health	Verify GET /vehicles and computed component health status	2 vehicles loaded. Chassis RB20-02 roll-up computed	PASSED
TC-TM-006	Vehicle Pairing History	Verify GET /vehicles/{id}/pairing-history chronological records	3 assignment history records retrieved for vehicle RB20-02	PASSED
TC-TM-007	Assignment & Health Guardrails	Verify POST /assignments, notification, audit, & Critical vehicle check	Created assignment; notification sent; HTTP 400 when Critical	PASSED
TC-TM-008	Driver Unassignment	Verify DELETE /assignments/{id} status deactivation & timestamps	Status updated to inactive; unassigned_at timestamp set	PASSED
TC-TM-009	Team Report Snapshotting	Verify POST /reports and GET /reports team-scoped snapshotting	Report generated with complete state; listed in history	PASSED
TC-TM-010	Race Calendar & Season Stats	Verify GET /calendar session driver filtering & GET /season-comparison	2023 calendar (22 events, filtered drivers); 2023 vs 2024 comparison	PASSED

3. FEATURE ARCHITECTURE & GUARDRAIL VERIFICATION

Driver-Vehicle Assignment & Mechanic Guardrail: Pairing history is maintained using the DriverVehicleAssignment table with partial unique database indexes. Prior to activating any assignment, the service queries get_vehicle_health(). If any component belonging to the target chassis has a status of critical, assignment is blocked immediately with an **HTTP 400 Bad Request**.

User & Staff Tenure Management (team_since): Following Alembic migration 0011_user_team_since, tenure dates are anchored to the users table. Endpoints PATCH /drivers/{id} and PATCH /staff/{id} correctly update user.team_since and emit structured audit events.

Race Calendar Telemetry Filtering: GET /calendar reuses the shared FastF1TelemetryProvider. Completed race results are filtered per session to contain *only* drivers who actually competed for the team during that race (max 2 drivers per session), preventing static team roster leakage.

4. RESOLVED ISSUES DURING TESTING

Issue: Response builders in team_manager.py threw an AttributeError when dereferencing driver.team_since instead of driver.user.team_since.
Fix: Updated lines 538, 742, 856, 948 in team_manager.py to safely dereference driver.user.team_since. All test cases subsequently passed cleanly.

5. TEST EXECUTION LOG CONSOLE OUTPUT

```
=====
TEAM MANAGER MODULE TEST EXECUTION SUMMARY
=====
Total Tests Executed : 10 | Passed : 10 | Failed : 0 | Pass Rate : 100%
-----
[PASS] TC-TM-001 | Team Manager Resolution & Scoping (Daril Tom Jose / Red Bull)
[PASS] TC-TM-002 | Dashboard Summary & Audit Feed (5 Drivers, 2 Vehicles, 15 Audits)
[PASS] TC-TM-003 | Driver Roster Management (Yuki Tsunoda #22 updated & restored)
[PASS] TC-TM-004 | Staff Directory & Tenure Management (12 Staff fetched, Isack Hadjar updated)
[PASS] TC-TM-005 | Vehicle Inventory & Health Rollup (Chassis RB20-02 roll-up computed)
[PASS] TC-TM-006 | Vehicle Pairing History Query (3 assignment records retrieved)
[PASS] TC-TM-007 | Assignment & Health Guardrails (HTTP 400 on Critical health verified)
[PASS] TC-TM-008 | Driver Unassignment Workflow (Status inactive, timestamp set)
[PASS] TC-TM-009 | Team Report Generation & Scoping (Generated report e5a40af3)
[PASS] TC-TM-010 | Race Calendar & Season Comparison (2023 vs 2024 stats verified)
=====
```